

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 2.3: EXCRETION IN PLANTS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This document shows you what a strong Final Explanation looks like. Use it as a model to answer our driving question: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Write your own answer in your own words, using evidence from your portfolio (interviews, photographs) and your group poster. Refer to the Acacia and Euphorbia sap, the droplets on grass and arrowroot leaves, and the falling yellow leaves. Each section below has a prompt to guide you, followed by an exemplar to learn from. Do not copy the exemplar. Use examples from your own community.

Part 1: What is excretion in plants, and why do plants have no kidneys?

Define excretion in plants. Explain why plants can manage without kidneys, bladders or other excretory organs. Give at least three reasons.

Excretion in plants is the removal of metabolic waste products, toxic substances and excess materials from plant tissues. You and I have kidneys, but a maize plant in a farm in Kericho or Nakuru does not need them. First, plants have a lower metabolic rate than animals, so they make less waste. Second, they produce less toxic nitrogen waste, because they reuse nitrogen to build proteins. Third, they recycle by-products. Oxygen from photosynthesis is used in respiration, and carbon dioxide from respiration is used in photosynthesis, so these gases are not just waste. Fourth, plants store insoluble, harmless wastes inside non-vital cells, such as dead cells in bark and old leaves, where they cannot hurt the plant. This is why the Acacia tree outside our school can survive with no organs for excretion.

Part 2: How do plants get rid of wastes?

Describe at least four ways plants eliminate wastes. For each, name the waste, the plant structure involved, and link it to something you observed (the sap, the leaf droplets, the falling leaves).

Plants use several methods. (1) Diffusion of gases: oxygen from photosynthesis and carbon dioxide from respiration move in and out through stomata on leaves and lenticels on stems. (2) Transpiration: extra water vapour leaves through the stomata. (3) Guttation: on the humid early mornings we saw droplets along the edges of grass and arrowroot (taro) leaves. When the soil is moist and the air is humid, transpiration is slow and root pressure pushes water up. The water, with dissolved salts, is forced out through special pores called hydathodes. (4) Exudation: when we cut an Acacia, gum oozed out and hardened, and the Euphorbia gave a milky latex. These gums, resins and latex are waste substances stored in or released from bark tissues, and they are released when the plant is damaged. (5) Deposition and shedding: insoluble wastes like calcium oxalate are stored in leaves, bark and fruits. When old yellowing leaves fall, or bark peels off, the plant gets rid of these wastes together with the tissue. So the

	oozing liquids and falling leaves we saw really are waste removal.
--	--

Part 3: Why are plant excretory products valuable?

Explain how humans use plant excretory products. Give at least three examples, using evidence from your community interviews and photographs. Include one example that needs careful regulation.	Many plant wastes are useful and are sold in Kenyan markets. Latex from rubber trees is used to make rubber for tyres and gloves. Tannins from tree bark are used to tan animal hides into leather, which is used for shoes and belts made by artisans in Nairobi and Kisumu. Gums such as gum arabic from Acacia trees in dry areas are used as adhesives and as thickeners in foods and drinks. Alkaloids are used in medicines. Quinine from cinchona bark treats malaria, an important disease around Lake Victoria and the coast. Caffeine is found in tea from Kericho and coffee from Kiambu. In my interview, a herbalist told me about plants used for treating coughs and wounds (use your own interview here). However, some alkaloids, like morphine, are habit-forming and can lead to substance abuse, so they must be used only under regulation, such as by a doctor's prescription. Because these products are made by plants for free as waste, but are useful to us, farmers and traders can earn money from them.
---	---

Part 4: Putting it together to answer the driving question

In one connected paragraph, answer the driving question. Connect how plants remove wastes, why they need no excretory organs, and why the wastes are valuable. Refer to your portfolio and group poster.	Trees and crops clear out their wastes without organs because their wastes are fewer, less toxic, often recycled, and can be stored safely in non-living parts. They release gases through stomata and lenticels, lose water vapour by transpiration, push out water and salts by guttation through hydathodes, ooze gums, resins and latex from bark, and store insoluble wastes like calcium oxalate in leaves and bark that are later shed. The sticky sap from the Acacia and Euphorbia, the droplets on arrowroot leaves and the falling yellow leaves that puzzled us at the beginning are all examples. These wastes are worth money because humans found uses for them: latex for rubber, tannins for leather, gums as adhesives and food thickeners, and alkaloids such as quinine and caffeine for medicine and drinks. Our portfolio photographs and interviews showed us that people in our community already collect and sell some of these. Finally, we must use habit-forming alkaloids carefully and follow the law to prevent abuse.
---	---

Part 5: Reflect on your evidence and the poster

Explain how one photograph or interview from your portfolio supports your explanation, and what your group poster shows. Say what you would still like to investigate.	One photograph in my portfolio shows an Acacia trunk with hardened gum. It supports my explanation because it shows exudation, a way plants get rid of waste, and the gum is later collected for sale. In an interview, a trader explained how she sells gum and tannin-rich bark (use your own example here). Our group poster used a labelled drawing of a leaf showing stomata and hydathodes, a picture of sap from a cut stem, and a table linking each product to its use and value. I still wonder whether the amount of gum an Acacia produces changes during a dry season, and how farmers can harvest it without harming the tree.
---	--

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of excretion in plants	Defines excretion correctly and gives four or more accurate reasons why plants lack excretory organs, plus correct mechanisms (gas diffusion, transpiration,	Defines excretion and gives two or three reasons and mechanisms with only minor errors.	Definition is unclear or incorrect, and few or inaccurate reasons and mechanisms are given.

	guttation, exudation, shedding of organs).		
Use of evidence from the phenomenon and portfolio	Links the Acacia and Euphorbia sap, leaf droplets and falling leaves, and at least two portfolio photographs or interviews, to the explanation.	Refers to the phenomenon and one portfolio item, but links are not always explained.	Little or no reference to the phenomenon or portfolio evidence.
Economic and medicinal applications	Explains at least three uses (e.g. latex, tannins, gums, alkaloids) with correct links to waste products and mentions regulated use of habit-forming alkaloids.	Gives two or three uses with mostly correct links; regulation is mentioned briefly or missed.	Gives one use or incorrect uses, with no link to excretory products.
Communication and Kenyan context	Clear, logical, well-organised writing with correct scientific terms and relevant Kenyan examples and places; the poster supports the writing.	Mostly clear writing with some scientific terms and some Kenyan examples.	Writing is hard to follow, with few scientific terms and no local examples.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.